

Lecture 5.4: Hand Lay-up and Spray Lay-up

Hand lay-up technique

Hand lay-up technique is the simplest method of composite processing. The infrastructural requirement for this method is also minimal. The processing steps are quite simple. First of all, a release gel is sprayed on the mold surface to avoid the sticking of polymer to the surface. Thin plastic sheets are used at the top and bottom of the mold plate to get good surface finish of the product. Reinforcement in the form of woven mats or chopped strand mats are cut as per the mold size and placed at the surface of mold after perspex sheet. Then thermosetting polymer in liquid form is mixed thoroughly in suitable proportion with a prescribed hardner (curing agent) and poured onto the surface of mat already placed in the mold. The polymer is uniformly spread with the help of brush. Second layer of mat is then placed on the polymer surface and a roller is moved with a mild pressure on the mat-polymer layer to remove any air trapped as well as the excess polymer present. The process is repeated for each layer of polymer and mat, till the required layers are stacked. After placing the plastic sheet, release gel is sprayed on the inner surface of the top mold plate which is then kept on the stacked layers and the pressure is applied. After curing either at room temperature or at some specific temperature, mold is opened and the developed composite part is taken out and further processed. The schematic of hand lay-up is shown in figure 1. The time of curing depends on type of polymer used for composite processing. For example, for epoxy based system, normal curing time at room temperature is 24-48 hours. This method is mainly suitable for thermosetting polymer based composites. Capital and infrastructural requirement is less as compared to other methods. Production rate is less and high volume fraction of reinforcement is difficult to achieve in the processed composites. Hand lay-up method finds application in many areas like aircraft components, automotive parts, boat hulls, diase board, deck etc. Generally, the materials used to develop composites through hand lay-up method are given in table 1.

Table 1 Raw materials used in hand lay-up method

Materials used	
Matrix	Epoxy, polyester, polyvinyl ester, phenolic resin, unsaturated polyester, polyurethane resin
Reinforcement	Glass fiber, carbon fiber, aramid fiber, natural plant fibers (sisal, banana, nettle, hemp, flax etc.) (all these fibers are in the form of unidirectional mat, bidirectional (woven) mat, stitched into a fabric form, mat of randomly oriented fibers)

Hand Lay-Up

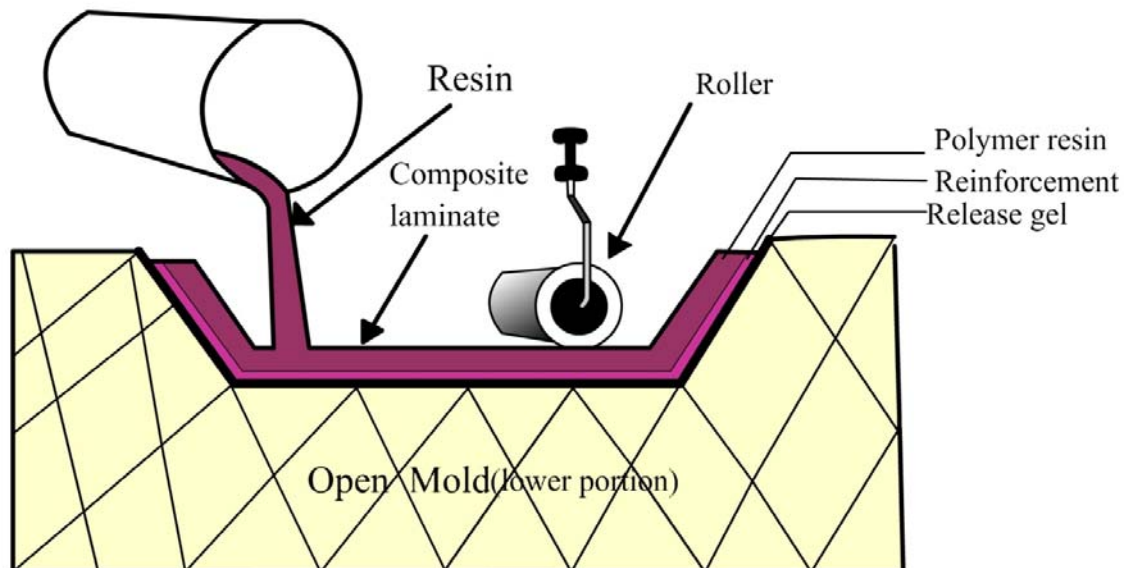


Figure 1 Hand lay-up method.

Spray lay-up

The spray lay-up technique can be said to be an extension of the hand lay-up method. In this technique, a spray gun is used to spray pressurized resin and reinforcement which is in the form of chopped fibers. Generally, glass roving is used as a reinforcement which passes through spray gun where it is chopped with a chopper gun. Matrix material and reinforcement may be sprayed simultaneously or separately one after one. Spray release gel is applied on to the mold surface to facilitate the easy removal of component from the mold. A roller is rolled over the sprayed material to remove air trapped into the lay-ups. After spraying fiber and

resin to required thickness, curing of the product is done either at room temperature or at elevated temperature. After curing, mold is opened and the developed composite part is taken out and further processed further. The time of curing depends on type of polymer used for composite processing. The schematic of the spray lay-up process is shown in figure 2. Spray lay-up method is used for lower load carrying parts like small boats, bath tubs, fairing of trucks etc. This method provides high volume fraction of reinforcement in composites and virtually, there is no part size limitation in this technique. Generally, the materials used to develop composites through spray lay-up method are given in table 2.

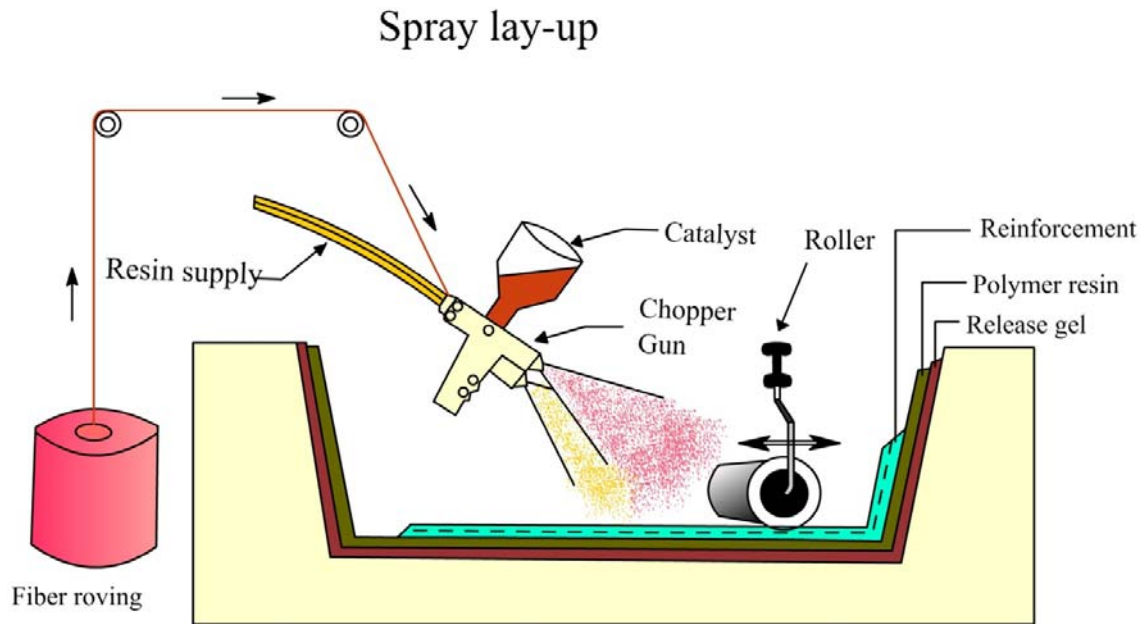


Figure 2 Spray lay-up method.

Table 2 Raw materials used in spray lay-up method

Materials used	
Matrix	Epoxy, polyester, polyvinyl ester, phenolic resin, unsaturated polyester, polyurethane resin
Reinforcement	Glass fiber, carbon fiber, aramid fiber, natural plant fibers (sisal, banana, nettle, hemp, flax, coir, cotton, jute etc.) (all these fibers are in the form of chopped short fibers, flakes, particle fillers etc.)

Lecture 5.10: Resin Transfer Molding and Autoclave Molding

Resin transfer molding

Resin transfer molding is a closed molding process. In this technique, as the name indicates, resin is transferred over the already placed reinforcement. Reinforcement in terms of either woven mat or strand mat form is placed on the surface of lower half mold. A release gel is applied on the mold surface for easy removal of the composite. The mold is properly closed and clamped. The clamping can be done either perimeter clamping or press clamping mechanism. The resin is pumped into the mold through ports and air is displaced through other vents. The uniformity of resin flow can be enhanced by using a catalyst as an accelerator and vacuum application. After curing, the mold is opened and composite product is taken out. The schematic of resin transfer molding process is shown in figure 1. Resin transfer molding can incorporate soft or hard mold depending upon the expected duration of run. For soft mold, thermosetting polymers like epoxy and polyester can be used for molding material. For hard mold, materials like steel and aluminium can be used. The cost of mold varies from very low to high cost mold with short to long life molds. The process can be automated to reduce cycle time. For complex shapes to be produced, preformed fiber reinforcements are used. The viscosity of the resin plays an important role in resin transfer molding process because injection time depends upon viscosity of the resin. If viscosity of resin is high, high pressure is required which may cause displacement of fibers, known as fiber wash.

The raw materials used in resin transfer molding are given in table 1.

Table 1 Raw materials used in resin transfer molding

Materials used	
Matrix	Epoxy, Methyl Methacrylate, polyester, polyvinyl ester, phenolic resin
Reinforcement	Glass fiber, carbon fiber, aramid fiber, natural plant fibers (sisal, banana, nettle, hemp, flax etc.) These fibers are used either individually or in combined form as a woven mat, unidirectional mat or chopped strand mat. Mineral fillers may also be added to improve surface finish and fire retardancy

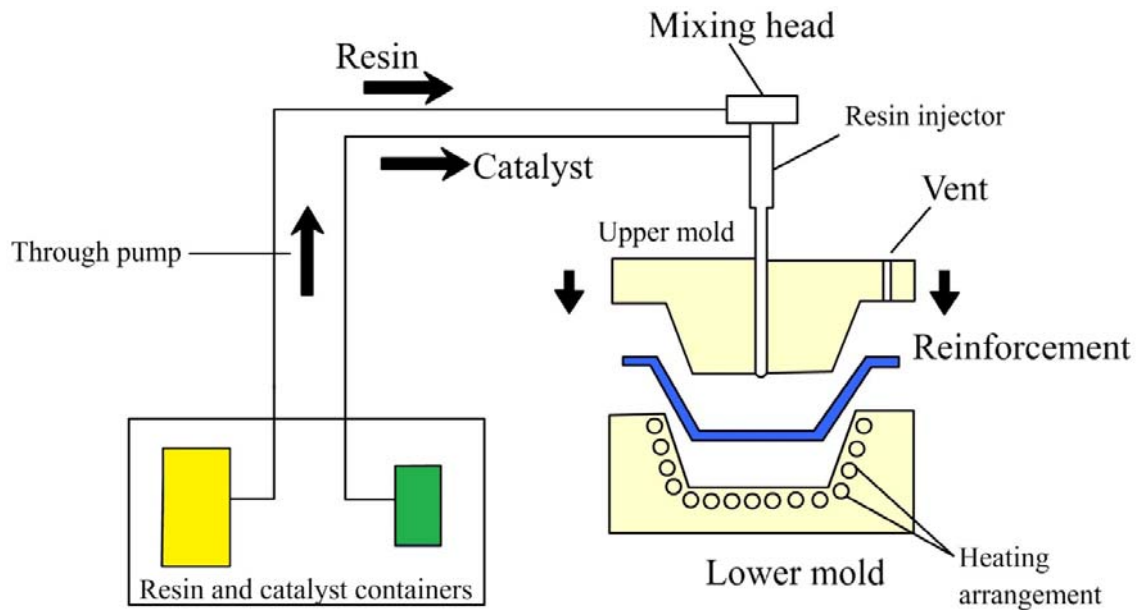


Figure1 Resin transfer molding.

Important components of resin transfer molding process:

There are mainly five components in the resin transfer molding system which govern the processing of composites. These components are:

1. Resin and catalyst container
2. Pumping unit
3. Mixing chamber
4. Resin injector
5. Molding unit

There are two separate containers for resin and catalyst. Resin container is larger than the catalyst container. Both the containers have separate outlets which pass through pumping unit and opens in mixing chamber. Pumping unit transfers the resin and catalyst to the mixing chamber. Resin and catalyst is mixed thoroughly in the mixing chamber. Resin injector is used to inject the mixture to the mold cavity. Molding unit has two halves namely upper half mold and lower half mold. Heating arrangement is integrated with molding unit. Vents are provided to release the gases in the mold cavity during clamping.

Application:

1. Hollow shapes and complex structure can be produced.

2. Automotive body parts, big containers, bathtubs are commonly processed through resin transfer molding technique.

Advantages:

1. Composite part produced with this method has good surface finish on both side surface of the product.
2. Any combination of reinforced materials (including 3D) in any orientation can be achieved.
3. Fast cycle time can be achieved through temperature control tooling device.
4. Process can be manual control, semi-automated or highly automated.
5. Composite part thickness is uniform which is determined by the mold cavity.
6. There is low emission during composite processing.
7. Strict dimensional tolerances are possible to achieve.
8. Ability to incorporate inserts and other attachments into mold.
9. The process does not require high injection pressure.
10. Material wastage is reduced as near net shape parts are produced.
11. Higher production rate is associated with process automation.

Disadvantages:

1. Mold cavity limits the size of the composite.
2. High tooling cost.
3. There is limitation on reinforcing materials due to the flow and resin saturation of fibers.

Vacuum Bag Molding

In vacuum bag molding, vacuum is created to remove entrapped air, gases and excess resin. As the lay-up of reinforcement (it may be a woven mat or other fabric form) and resin is completed then a non-adhering film of nylon or polyvinyl alcohol (PVA) is placed over the lay-up and sealed. These films forms a bag through which vacuum is created within the mold and at this condition composites are cured either at room temperature or at any specific temperature. In this process, atmospheric pressure is used to suck air under vacuum bag which compacts composite layers down and produces a superior quality laminate.

Pressure Bag Molding

Pressure bag molding process which is same as the vacuum bag molding process with the only difference of air pressure. Air pressure is applied to eliminate entrapped gases and excess resin to the film or bag of poly vinyl alcohol or nylon which covers the lay-ups of fiber and resin matrix. Sometimes, pressurized steam is also used instead of air which has dual benefits. Steam removes excess air as well as provides curing to the composite part.

Autoclave molding

Autoclave molding technique is similar to vacuum bag and pressure bag molding method with some modifications. This method employs an autoclave to provide heat and pressure to the composite product during curing. In this method, prepregs are stacked in a mold in a definite sequence and then spot welded to avoid any relative movement in between the prepreg sheets. After stacking the prepregs, the whole assembly is vacuum bagged to remove any air entrapped in between the layers. The schematic of autoclave molding process is shown in figure 2. After a definite period of time when it is ensured that all air is removed, the entire assembly is transferred to autoclave. Here, heat and pressure is applied for a definite interval of time. In this process, matrix is uniformly distributed and intimate contact is achieved through proper bonding between fibers and matrix. After the processing, the assembly is cooled to a definite rate and then vacuum bag is removed. The composite part is taken out from the mold. Initially, a release gel is applied onto the mold surface to avoid sticking of polymer to the mold surface. The raw materials used in these techniques are given in the table 1.

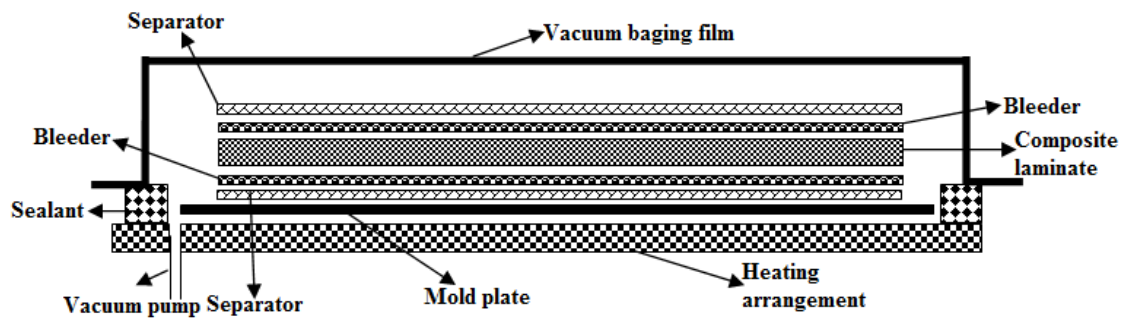


Figure 2 Autoclave molding process.

Table 1: Raw materials used in autoclave molding process

Materials used	
Matrix	Epoxy, polyester, polyvinyl ester, phenolic resin, Unsaturated polyester, polyurethane resin and thermoplastic resins,
Reinforcement	Glass fiber, carbon fiber, aramid fiber (all these fibers may be in the form of unidirectional mat, bidirectional (woven) mat, stitched into a fabric form, mat of randomly oriented fibers)

Application:

The process is mainly used in applications requiring high strength to weight ratio components such as aircraft parts, marine, military, space craft and missiles.

Advantages

1. This composite processing method allows high volume fraction of reinforcement in the composite part.
2. This method is applicable for both thermoplastic and thermosetting polymer composites.
3. High degree of uniformity in part consolidation, better adhesion characteristics between layers and good control over resin and reinforcement is achieved.
4. No void content in the finished part due to removing entrapped air through vacuum.
5. If cores and inserts are used, there is better bonding of these attachments due to vacuum bag processing.
6. Complete wetting of fibers is achieved.

Disadvantages

1. There is limitation on part size which depends upon autoclave size.
2. It is a costly technique for composite processing.
3. Rate of production is low and skilled labour is required in this process.